

POPRAD-TATRY, Slovakia

WMO: 119340

Lat: 49.074N

Lon: 20.241E

Elev: 2356

StdP: 13.49

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.2	5.1	-5.0	4.6	1.0	-0.2	5.9	7.0	30.9	34.0	28.1	34.1	3.3	270	0.612

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.1	82.6	65.9	79.2	64.2	76.4	62.9	67.4	79.5	65.7	76.9	64.1	74.2	8.8	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.0	94.1	72.7	61.4	88.9	70.7	60.0	84.5	69.2	33.2	79.8	31.9	77.1	30.6	74.3	73.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.9	21.7	18.9	DB	-6.7	87.9	6.2	2.9	-11.2	90.0	-14.8	91.8	-18.3	93.4	-22.8	95.6	(4)
(5)			WB	-7.3	70.5	6.1	1.8	-11.7	71.7	-15.2	72.8	-18.6	73.8	-23.1	75.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.4	24.6	27.8	35.0	45.1	53.3	59.3	62.5	61.9	53.6	45.3	36.4	26.9		
	DBStd	15.37	9.59	9.54	7.98	7.18	6.24	5.89	5.48	5.48	6.17	7.59	8.52	9.14		
	HDD50	3443	789	621	464	176	39	3	0	1	33	187	413	717		
	HDD65	7610	1254	1041	929	598	365	186	115	127	343	611	859	1182		
	CDD50	1395	0	0	1	27	140	283	387	371	142	40	4	0		
	CDD65	86	0	0	0	0	1	16	36	31	2	0	0	0		
	CDH74	1195	0	0	0	3	43	240	445	425	38	1	0	0		
	CDH80	236	0	0	0	0	3	39	95	95	4	0	0	0		
	WSAvg	7.3	7.5	7.7	8.7	7.7	7.4	7.1	7.2	6.3	6.7	6.8	7.1	7.1		
	PrecAvg	23.7	0.9	1.0	1.1	1.7	2.9	3.5	3.3	2.9	2.0	1.7	1.7	1.1		
(15) Precipitation	PrecMax	39.3	3.0	3.3	3.4	4.6	6.2	8.4	8.7	7.3	4.7	7.4	5.3	3.3		
	PrecMin	16.3	0.0	0.1	0.1	0.2	0.2	0.9	0.3	0.4	0.1	0.0	0.0	0.0		
	PrecStd	4.6	0.6	0.7	0.7	1.0	1.3	1.4	1.8	1.5	1.1	1.5	1.1	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.7	51.9	61.3	72.0	78.6	84.1	87.3	87.6	78.4	71.2	59.2	48.6		
		MCWB	41.2	43.1	46.3	55.0	60.6	66.6	68.2	66.3	62.6	57.1	51.3	44.1		
	2%	DB	43.0	46.8	55.8	67.6	74.1	80.3	82.4	82.4	73.6	66.2	55.1	44.8		
		MCWB	39.9	41.0	45.1	52.6	59.5	64.8	66.1	66.1	60.6	55.7	49.1	41.7		
	5%	DB	40.0	43.8	51.6	62.9	70.8	76.6	78.9	78.6	70.1	62.4	51.8	41.4		
		MCWB	37.1	38.8	43.2	50.3	57.8	63.1	64.5	64.2	58.5	54.0	47.2	38.7		
	10%	DB	37.3	40.7	47.5	59.0	66.9	73.0	75.4	75.2	66.4	58.1	48.6	38.7		
		MCWB	35.0	36.9	41.0	48.6	55.9	61.4	63.0	62.5	56.8	51.8	45.1	36.4		
	0.4%	WB	42.7	44.6	48.6	56.5	63.9	69.3	69.8	69.3	64.5	59.0	52.0	44.9		
		MCDB	44.9	49.5	57.3	68.8	74.8	79.9	80.9	81.4	75.8	67.0	57.2	47.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	40.2	41.8	46.1	53.8	60.8	66.3	67.4	67.1	62.0	56.5	49.5	42.0		
		MCDB	42.6	46.3	53.9	64.6	71.2	77.5	79.5	79.6	71.3	64.3	54.1	44.5		
	5%	WB	37.3	39.3	43.9	51.7	58.7	64.1	65.7	65.3	59.8	54.5	47.7	38.8		
		MCDB	39.6	42.8	50.3	61.1	68.6	74.2	76.5	76.6	68.1	61.4	51.5	40.8		
	10%	WB	35.0	37.0	41.4	49.4	56.7	62.3	64.0	63.5	57.8	52.3	45.4	36.6		
		MCDB	36.9	40.1	46.7	57.5	65.4	71.4	73.7	73.3	65.0	57.9	48.5	38.3		
	5% DB	MDBR	13.7	15.5	17.2	20.1	20.8	20.9	21.1	22.2	19.9	18.3	13.5	12.7		
		MCDBR	12.2	17.9	24.4	27.8	27.3	27.4	27.4	27.7	26.3	24.9	17.4	13.3		
		MCWBR	10.0	12.8	15.4	15.3	14.8	14.6	13.5	13.5	14.7	15.4	12.3	10.8		
		MCWBR	11.4	15.5	21.5	24.7	24.7	24.7	24.7	25.7	23.3	22.8	15.4	12.0		
(40) Clear-Sky Solar Irradiance	taub		0.288	0.315	0.362	0.424	0.418	0.420	0.434	0.430	0.396	0.360	0.328	0.288		
		taud	2.280	2.217	2.179	2.126	2.211	2.243	2.206	2.223	2.284	2.356	2.370	2.339		
	Ebn at Noon		241	260	266	260	267	267	261	256	253	242	225	226		
		Edh at Noon	23	31	38	45	43	42	43	41	35	27	21	19		
(44) All-Sky Solar Radiation	RadAvg		292	531	907	1312	1583	1693	1639	1509	1065	649	333	223		
	RadStd		41	81	128	144	195	145	178	129	149	91	44	25		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.81	N/A	N/A	N/A	+1.12	+0.97	N/A	N/A	+94	+27
(47)	Regional (45 neighbors)	+1.03	N/A	N/A	+1.71	+0.92	+0.81	-227	-326	+160	+72

Nomenclature: See separate page